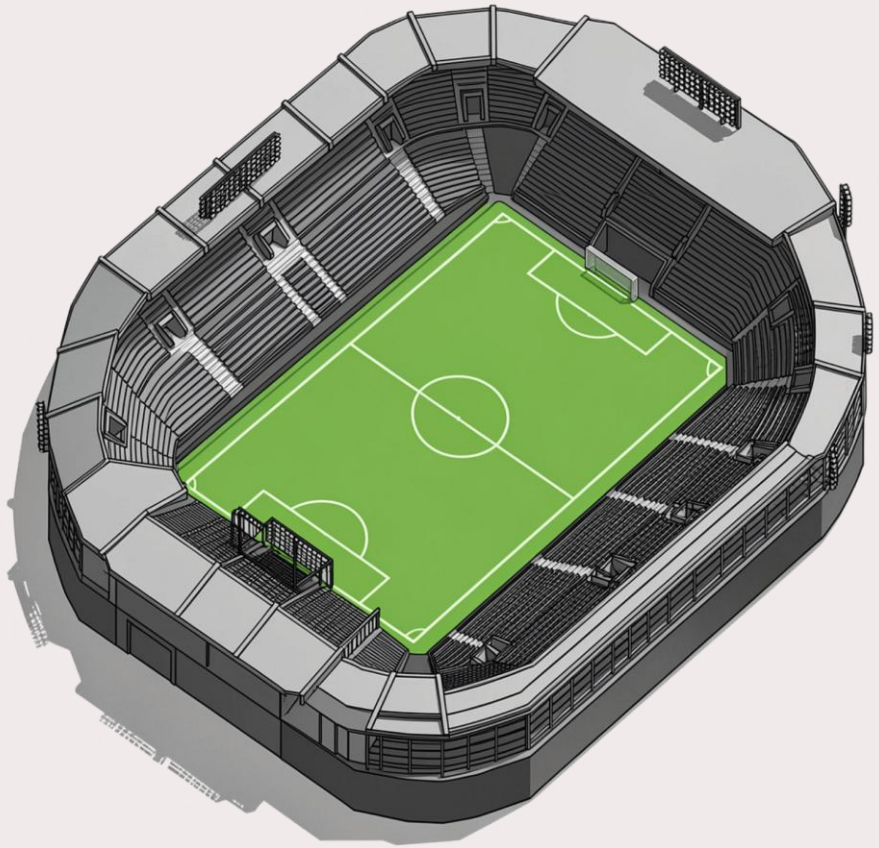


Sports Club Membership & Tournament Management System

DBMS Project | **Parnika Podili** (25WU0102194) | Faculty: Prof. Sahithi



The Problem: Manual Chaos

Fragmented Records

Member, team, coach, and facility data managed in silos — no centralized system.

Scheduling Conflicts

Training sessions and fixtures clash constantly without automated conflict detection.

Scattered Financials

Fee payments and equipment records are dispersed, making audits unreliable.



Why This Matters

- Generating reports and standings consumes excessive staff time
- Data inconsistencies trigger errors and member disputes
- Manual processes cap the club's capacity to grow

Solution needed: A unified DBMS to streamline all club operations end-to-end.

Project Objectives

Centralize Records

Members, teams, coaches, and facilities in one system.

Automate Scheduling

Eliminate conflicts across training and fixtures.

Track Tournaments

Real-time results, fixtures, and live standings.

Enforce Integrity

Data accuracy guaranteed via DBMS constraints.

System Scope: 8 Core Modules

1

Members & Memberships

Registration, profiles, status

2

Sports, Teams & Coaches

Composition & assignments

3

Facilities & Training

Venue booking & sessions

4

Tournaments & Fixtures

Events & match scheduling

1

Results & Standings

Outcomes & rankings

2

Fees & Payments

Dues & payment tracking

3

Equipment

Inventory, issuance & returns

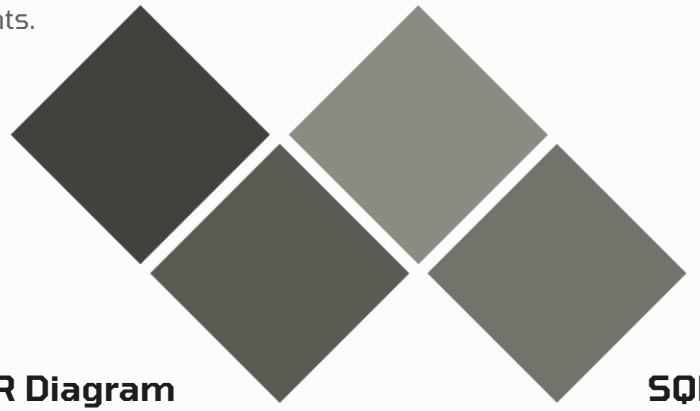
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Reports

Dynamic reports & analytics

Technical Architecture

- **Database:** MySQL with a fully relational schema normalized to 3NF
- **Frontend:** Python application with a user-friendly interface
- **Design approach:** ER Diagram → Relational Schema → Normalized Tables → SQL Implementation



Relational Schema

Map entities to tables with keys and constraints.

3NF Normalization

Eliminate redundancies and ensure dependency rules.

ER Diagram

Model entities, attributes, and relationships visually.

SQL Implementation

Create MySQL tables, indexes, and integrity rules.

Development Roadmap

1. Problem Identification & Requirements Analysis
2. ER Diagram & Database Design
3. Relational Schema & 3NF Normalization
4. SQL Tables, Constraints & Sample Data
5. Queries, Views & Reports
6. Python Frontend & System Testing

Functional Requirements

CRUD Operations

Create, read, update, and delete records for members, teams, facilities, and equipment.

Automated Scheduling

Conflict detection across training sessions and tournament fixtures.

Payment Recording

Fee management with complete payment history and audit trails.

Dynamic Reports

Real-time report generation for performance, finances, and operations.

Tournament Management

End-to-end fixture creation, result tracking, and standings calculation.

System Users & Access Levels



Administrator

Full control — members, teams, facilities, and all system records.



Coach

Manages team rosters and training session scheduling.



Staff

Handles fixtures, payment processing, and equipment records.



Member

View-only access to personal profile and tournament information.



Expected Outcome

Unified System

Centralized, easy-to-use DBMS eliminating all manual processes.

Conflict-Free Operations

Automated scheduling with real-time conflict detection.

Data Integrity

Accurate, auditable records with full transaction history.

Scalable Growth

A robust foundation ready to support the club's future expansion.